Chapter 18, Problem 57P

Problem

Let $i(t) = 2etu(-t)$ A. Find the total energy carried by i(t) and the percentage of the 1-**Ω** energy in the frequency range of $-5<\omega<5$ rad/s.

Step-by-step solution

Step 1 of 1

$$W_{1\Omega} = \int_{-\infty}^{\infty} I^2(t)\,dt = \int_{-\infty}^{0} 4e^{2t}\,dt$$

$$= 2e^{2t}\int_{-\infty}^{0} = 2j$$

$$I(\omega) = \frac{2}{1-j\omega} \,|\, I(\omega)|^2 = \frac{4}{1+\omega^2}$$

$$W_{1\Omega} = \frac{1}{2\pi}\int_{-\infty}^{\infty} |I(\omega)|^2\,d\omega = \frac{4}{2\pi}\int_{-\infty}^{\infty} \frac{1}{1+\omega^2}\,d\omega = \frac{4}{\pi}\tan^{-1}(\omega)\Big|_0^{\infty} = \frac{4}{\pi}.\frac{\pi}{2} = 2J$$

In the frequency range

$$W = \frac{4}{\pi}\tan^{-1}\omega\Big|_0^{5} = \frac{4}{\pi}(1.373) = 1.7487$$

$$\frac{W}{W_{1\Omega}} = \frac{1.7467}{2} = 0.8743\%$$